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Effects of Credit Expansions on Stock Market Booms and Busts

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1 Big picture

- **Question.** Does a large expansion in margin credit raise stock prices? The theoretical answer is not obvious because equity markets are much more liquid than housing markets, and relatively unconstrained arbitrageurs may offset the demand of newly levered retail traders.
- **Setting.** The paper studies China's margin-lending deregulation between 2010 and 2015. Regulators gradually rolled out margin eligibility across stocks, first through pilots and then through three formula-based vintages. This gives the authors both staggered event-time variation and a threshold-based regression discontinuity design.
- **Main challenge.** A simple before-versus-after comparison is not enough. If the market could foresee which stocks would become marginable, prices could rise *before* the formal rollout. Then the absence of a jump at implementation would not imply that credit had no effect — it could mean the effect was already priced in.
- **Main empirical contribution.** The paper combines three pieces of evidence:
 - event studies showing strong pre-marginability returns and almost no average post-marginability jump,
 - regression discontinuity evidence around the formula-based threshold showing a sharp *causal* increase in prices for barely qualifying stocks,
 - mutual-fund and large-holder evidence showing that relatively unconstrained investors bought ahead of the rollout and sold afterward.
- **Main finding.** Margin lending had a large positive causal effect on stock prices, but much of that effect was anticipated and front-run by sophisticated investors. In the Chinese setting, deep-pocketed investors acted more like front-runners than contrarians.
- **Magnitude.** The regression discontinuity estimates imply that becoming marginable raised cumulative returns by roughly 5% after 5 trading days, 8.5% after 20 trading days, and about 25% after 60 trading days. That is a very large demand effect for a market-level credit expansion.
- **Interpretation of the event-study patterns.** The paper's most important empirical point is the combination of:
 - near-zero abnormal returns *after* a stock becomes marginable on average,
 - strong abnormal returns *before* the rollout,
 - and a large causal RD effect right at the margin of surprise.

Together these patterns are exactly what one would expect if the market anticipated a real credit-supply shock and impounded it into prices before the official implementation date.

- **Institutional-investor evidence.** Mutual funds and other large owners gradually increased their holdings of soon-to-be-marginable stocks before the deregulation and then reduced those positions afterward. The funds' anticipated positions earned economically meaningful abnormal returns.
- **Model contribution.** The paper develops a dynamic stock-pricing model with information revelation. Investors gradually receive signals about the size of the coming credit expansion.

Under an exponential information structure, the model generates exponential pre-trends in prices and fits the data well with only two key parameters: the direct effect of the credit shock and the speed of anticipation.

- **Model estimate.** The paper estimates an effective monthly anticipation rate of roughly 5% to 10%, with more than 60% of the ultimate price effect already priced in six months before the event in the preferred estimates.
- **Broader implication.** The paper does not claim a clean causal estimate for the entire 2015 Chinese boom-bust cycle, but the back-of-the-envelope calculations suggest that margin debt plausibly explains about one-fifth of the average boom and that stocks with more margin debt at the peak crashed more severely.
- **Economic takeaway.** Large shocks to credit for stock purchases can push up equity prices substantially. The reason the effect can be hard to see in simple event windows is that sophisticated traders may anticipate it and buy early.

2 Data and measurement (key objects)

2.1 The deregulation setting

The Chinese government gradually liberalized margin lending for stocks listed on the Shanghai and Shenzhen exchanges between 2010 and 2015.

The reform had two phases:

- **Pilot phase.** In Pilot A, announced on February 13, 2010, 50 Shanghai stocks and 40 Shenzhen stocks became marginable. In Pilot B, announced on November 25, 2011, an additional 131 Shanghai stocks and 60 Shenzhen stocks were added.
- **Formula-based phase.** Three later vintages were selected by a published ranking formula based on tradable market value and trading volume.

By Vintage 3, roughly 900 stocks in total could be purchased on margin. The share of floating market capitalization that was marginable rose from 53.05% in Pilot A to 79.30% by Vintage 3.

2.2 Who could use margin and at what terms

Retail investors qualified for brokerage-provided margin lending if they had:

- at least 500,000 RMB in brokerage assets,
- at least six months of trading experience,
- and they faced an initial margin requirement of 50%.

Institutional investors were not formally allowed to use broker margin during most of the sample. The government actively supported the reform, including through China Securities Finance Co., which funded brokerages.

Formal margin interest rates were around 8% to 9% annualized, much lower than the roughly 25%

rate on shadow margin loans that became important only later.

2.3 The inclusion index and marginability rule

For the formula-based vintages, regulators first screened out stocks that were too small, too new, too illiquid, too volatile, or otherwise ineligible. Among the remaining stocks, they ranked candidates using the inclusion index:

$$\text{Inclusion Index}_i = \theta \times \frac{\text{Average tradable market value of stock } i}{\text{Average tradable market value of all stocks in SH/SZ}} + \gamma \times \frac{\text{Average trading volume in yuan of stock } i}{\text{Average trading volume of all stocks in SH/SZ}} \quad (1)$$

Interpretation:

- the first term rewards size,
- the second term rewards liquidity,
- and the threshold is the inclusion-index value of the last stock that makes the vintage.

For Shenzhen, the exchange used $\theta = 2$ and $\gamma = 1$ in all three vintages. For Shanghai, the exchange used $\theta = \gamma = 1$ for the first two vintages and then switched to $\theta = 2$ and $\gamma = 1$ for Vintage 3.

2.4 Timeline and coverage by vintage

Table 1 in the paper reports:

- **Pilot A** (February 13, 2010): 50 Shanghai and 40 Shenzhen stocks; 53.05% of floating cap.
- **Pilot B** (November 25, 2011): 131 Shanghai and 60 Shenzhen stocks; 65.86%.
- **Vintage 1** (January 25, 2013 announcement; January 31 implementation): 163 Shanghai and 113 Shenzhen stocks; 75.94%.
- **Vintage 2** (September 6, 2013 announcement; September 16 implementation): 104 Shanghai and 102 Shenzhen stocks; 78.22%.
- **Vintage 3** (September 12, 2014 announcement; September 22 implementation): 104 Shanghai and 114 Shenzhen stocks; 79.30%.

2.5 Formal margin, short selling, and shadow margin

The paper emphasizes three market-design facts:

- **Formal margin debt expanded enormously.** Aggregate broker-provided margin debt rose from essentially zero to about 4.5% of floating market capitalization in the overall market, and reached 8% to 10% for many treated vintages.
- **Short selling was negligible.** Aggregate short interest peaked at only 0.02% of floating market capitalization in 2013, so the liberalization was effectively about leveraged buying rather than balanced long/short access.

- **Shadow margin became important only later.** The paper estimates that shadow margin was virtually nonexistent before 2014, was tiny relative to formal margin in early 2014 (about 90 million yuan versus more than 300 billion yuan in formal margin), and was still less than one-fifth of formal margin at the time of Vintage 3.

Interpretation. These facts are important because the reduced-form analysis of the initial rollout is not contaminated by a large securities-lending market or by a large pre-existing shadow-margin system.

2.6 Mutual fund and ownership data

The paper uses several ownership datasets:

- full equity holdings of Chinese mutual funds at semiannual frequency,
- quarterly top-10 holdings for each mutual fund,
- quarterly stock-level ownership shares of the largest 10 investors in each stock.

Mutual funds held only 4% to 5% of the Chinese stock market in this period, far less than in the United States, but they were active and speculative enough to serve as the paper's main proxy for relatively unconstrained arbitrageurs.

2.7 Core data sources

The paper uses:

- stock price, trading, and financial data from CSMAR,
- stock-level daily margin debt from the Shanghai and Shenzhen exchanges,
- mutual fund holdings data,
- and proprietary stock-level shadow margin data from a provider that covers about 5% of the market.

The main analysis period for the reduced-form exercises is March 2009 to May 2015.

2.8 Main outcome variables

The paper studies several distinct objects:

- **marginability:** whether a stock becomes eligible for margin lending,
- **stock-level margin debt:** the quantity of margin debt outstanding,
- **margin debt / market cap:** leverage relative to floating market capitalization,
- **returns:** DGTW-adjusted cumulative returns in the event study and raw cumulative returns in RD,
- **institutional ownership shares:** for mutual funds and top-10 investors,

- **mutual-fund portfolio weights:** to test anticipatory buying,
- **crash return:** the log return during the 2015 bust window,
- **peak total margin:** formal plus shadow margin debt relative to floating market capitalization at the June 2015 peak.

2.9 Control groups

Different parts of the paper use different counterfactual groups:

- **Never-marginable stocks** are the main control group in event studies and in the dynamic panel model.
- **Stocks just below the threshold** are the control group in the regression discontinuity design.
- **Non-TBM holdings** within mutual fund portfolios are the control holdings in the alpha tests.

Interpretation. The paper is careful to match the identifying strategy to the empirical question. Broad controls are used for anticipation patterns; local threshold comparisons are used for causal identification.

3 Models and empirical specifications (all equations + interpretation)

3.1 Baseline cross-sectional event-study specification

For each vintage k , the paper compares stocks that become marginable in that vintage to stocks that never become marginable:

$$\text{Ret}_i^k = \beta_0 \text{Marginable}_i^k + \theta_k + \varepsilon_i^k. \quad (2)$$

Here:

- Ret_i^k is the cumulative DGTW-adjusted return for stock i in a window around the implementation month for vintage k ,
- Marginable_i^k equals 1 if stock i becomes marginable in vintage k ,
- θ_k is a vintage fixed effect.

Interpretation. β_0 measures whether treated stocks outperform never-marginable stocks in a chosen horizon. The paper runs this both *after* marginability and *before* marginability. The striking finding is that the post window is close to zero on average, whereas the pre window is strongly positive.

3.2 Stacked event-study specification

To trace a long path of anticipatory price changes, the paper stacks vintages and estimates:

$$\text{Ret}_{ikt} = \sum_{\ell} \beta_{\ell} 1\{t - \tau(k) = \ell\} \times \text{Treated}_{ik} + \gamma_{ik} + \lambda_{tk} + \varepsilon_{ikt}. \quad (3)$$

Here:

- $\tau(k)$ is the implementation date of vintage k ,
- Treated_{ik} marks stocks that eventually become marginable in vintage k ,
- γ_{ik} are stock-vintage fixed effects,
- λ_{tk} are time-vintage fixed effects.

Interpretation. This gives a dynamic plot of treated-minus-control returns over a very long horizon. The paper finds a gradual, roughly exponential run-up before implementation.

3.3 Regression discontinuity design

The core RD equation is:

$$Y_i^k = \alpha_{0\ell} + \alpha_{1\ell}(\text{Index}_i^k - C_E^k) + \tau_i^k \left[\alpha_{0r} + \alpha_{1r}(\text{Index}_i^k - C_E^k) \right] + \theta_k + \lambda_{\text{ind}(i)} + \varepsilon_i^k. \quad (4)$$

Here:

- Index_i^k is the inclusion index,
- C_E^k is the exchange-specific threshold for vintage k ,
- τ_i^k indicates whether the stock is above the threshold,
- θ_k are vintage fixed effects,
- $\lambda_{\text{ind}(i)}$ are industry fixed effects.

Interpretation. α_{0r} is the discrete jump at the threshold. The paper first uses this as a first stage for marginability and margin debt, then as a reduced-form price effect, and finally in a fuzzy-RD setup where being above the threshold instruments for becoming marginable.

3.4 Mutual-fund anticipatory buying

At the fund-stock level, the paper estimates:

$$w_{ijt} = \alpha + \beta \text{TopRanking}_{it} \times \text{MonthsSinceLastVintage}_t + \text{Controls}_{it} + \sigma_i + \theta_j + \gamma_t + \varepsilon_{ijt}. \quad (5)$$

Here:

- w_{ijt} is stock i 's portfolio weight in fund j at date t ,

- TopRanking_{it} identifies stocks that appear likely to become marginable based on contemporaneously available index inputs,
- $\text{MonthsSinceLastVintage}_t$ captures how close the market is to the next likely rollout.

Interpretation. A positive β means funds tilt more toward likely-to-be-marginable stocks as the next vintage approaches.

3.5 Mutual-fund selling after qualification

To test whether funds unwind after the event, the paper estimates:

$$w_{ijt} = \alpha + \beta \text{Marginable}_{it} \times \text{MonthsSinceMarginable}_{it} + \text{Controls}_{it} + \sigma_i + \theta_j + \gamma_t + \varepsilon_{ijt}. \quad (6)$$

Interpretation. A negative β means mutual funds gradually sell newly marginable stocks after retail margin buyers arrive.

3.6 General information-revelation pricing model

The structural section introduces a competitive stock-pricing model with risk-averse unconstrained investors and a future credit shock at $t = 0$.

A reduced-form demand shock of Δ shares at $t = 0$ generates a direct price effect m , which can be interpreted as the causal impact of the credit expansion if the shock were unanticipated.

With gradual signals about the shock, the equilibrium price for $t \leq 0$ is:

$$p_t = p^* + \sum_{j=-n}^t m_j - \gamma \sum_{k=t+1}^0 \sigma_k^2 Q. \quad (7)$$

Interpretation. Price rises before the event for two reasons:

- investors learn about the future credit shock through the realized signals m_j ,
- and the risk discount falls as uncertainty over the size of the shock shrinks.

3.7 Exponential information structure

The paper imposes an exponential decay structure on the variance of the signals:

$$\sigma_t^2 = \beta \theta^t, \quad \theta > 0. \quad (2)$$

For ex post treated stocks ($m > 0$), expected prices follow:

$$E[p_t \mid m > 0] = \tilde{p} + \beta(\lambda + \gamma) \sum_{j=-\infty}^t 1\{j \leq 0\} \theta^j. \quad (8)$$

Interpretation. θ governs the speed of anticipation. Lower values imply earlier run-ups; values closer to 1 imply steep, late anticipation. The model implies that pre-event price increases follow an

exponential curve.

3.8 Nonlinear pre-trend estimation

Taking first differences yields:

$$p_{it}^{\text{treated}} - p_{i,t-1}^{\text{treated}} = \delta_1 \theta^t + \Delta \varepsilon_{it}, \quad t \leq 0. \quad (9)$$

Here:

- $\delta_1 = \beta(\lambda + \gamma)$ is the average price increase on the rollout date,
- θ again controls the speed of information revelation.

The total direct effect implied by the model is:

$$\Delta p_{-\infty} = \frac{\delta_1}{1 - 1/\theta}. \quad (3)$$

Interpretation. This converts the fitted one-period exponential increments into a total pre-event-plus-event price effect.

3.9 Linear dynamic panel representation

The model also implies:

$$p_{it} = \delta_1 D_{it} + \frac{1}{\theta} p_{i,t+1} + \alpha_i + \eta_t + e_{it}. \quad (10)$$

Here D_{it} equals 1 only for treated stocks after margin becomes active.

Interpretation. This is a forward-looking dynamic panel equation. OLS is biased because $p_{i,t+1}$ is endogenous, so the paper uses a 2SLS approach with leads of treatment as instruments.

3.10 Back-of-the-envelope demand elasticity

To connect the reduced-form price effect to market-level demand elasticity, the paper uses:

$$\text{Return} = -\frac{1}{\varepsilon} \% \Delta \text{Demand}. \quad (11)$$

Interpretation. Here ε is the elasticity of demand of unconstrained investors. Using the RD estimate of a roughly 26% price effect and a roughly 5% increase in margin debt relative to market cap, the paper backs out $\varepsilon \approx -0.19$, implying inelastic demand in this market.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the baseline event-study results

The baseline event-study evidence is identified by comparing soon-to-be-marginable stocks to never-marginable stocks within vintages and over time. The paper's central insight is that the *absence* of a post-event average jump is not evidence against a credit effect once one sees large pre-event run-ups.

Interpretation. Event studies here are mainly informative about *anticipation patterns*, not by themselves about a clean causal treatment effect.

4.2 Why the event-study evidence alone is not enough

The authors are explicit that pre-trends can always be suspicious. A stock that is about to become marginable may also be on a different fundamental trajectory for unrelated reasons. That is why the paper does not stop at Table 2. It turns to a threshold-based design whose local variation is plausibly unanticipated.

4.3 Why the regression discontinuity design is credible

The RD design exploits a published formula that determines eligibility. This is powerful for three reasons:

- the ranking rule is known,
- the inputs (market value and turnover) move at high frequency,
- and the exact number of stocks included in a given vintage is not perfectly known in advance.

So for stocks very near the threshold, eligibility is meaningfully uncertain ex ante. A tiny move in trading volume or market capitalization can switch a stock from just below to just above the cutoff.

Interpretation. The RD compares very similar stocks in a neighborhood where treatment assignment is quasi-random from the standpoint of investors trying to forecast the final list.

4.4 No manipulation around the threshold

The paper runs McCrary-style density tests following Cattaneo, Jansson, and Ma (2018). Figure 3 shows no evidence of bunching around the threshold; the density-test t -statistic is just below 1. In addition, the paper reports no meaningful discontinuities in ex ante observables around the cutoff.

Interpretation. This matters because if insiders or market participants could systematically game the inclusion rule locally, the RD would be compromised. The paper finds little evidence of such gaming.

4.5 A very strong first stage

Crossing the threshold almost perfectly predicts treatment:

- the probability of becoming marginable jumps by about 0.96,

- stock-level margin debt rises by about 0.359 billion yuan,
- and margin debt rises by about 5% of floating market capitalization.

Interpretation. This is not a weak or fuzzy policy margin in the colloquial sense. The threshold creates a very large differential credit shock, which is why the causal price effects are so precisely estimated.

4.6 Why anticipation does not invalidate the RD

The authors argue that the RD sample is special: investors may have known the formula, but for stocks right at the cutoff they could not know with certainty which names would end up qualifying. Therefore, unlike the broader event-study sample, the RD neighborhood is much less contaminated by anticipation.

This is exactly why the paper combines both designs:

- the broad event study documents market-wide anticipation,
- the local RD isolates the surprise component near the threshold.

4.7 Institutional holdings as mechanism evidence

The institutional-holdings evidence is not the main identification device, but it is a major credibility device. It shows that the agents one would expect to front-run a predictable credit shock — mutual funds and large investors — actually did so.

Interpretation. This ties together the reduced-form facts: the pre-run-up is not a statistical accident; it lines up with who was buying and when.

4.8 Why short selling and shadow margin are not the main story

Two possible alternative channels are muted:

- short selling is too small to matter quantitatively,
- shadow margin is basically absent in the early vintages and still secondary at the time of Vintage 3.

Interpretation. The initial price effects are really about the introduction of formal leveraged buying capacity.

4.9 The dynamic model's identifying assumption

The dynamic panel model uses treated stocks and never-marginable stocks and assumes that, absent the deregulation, the timing of vintages is orthogonal to vintage-specific demand shocks for treated stocks.

The paper checks this in a difference-in-difference spirit by showing that long before meaningful anticipation begins, the price paths of treated and never-marginable stocks look similar.

4.10 What the paper can and cannot identify

The paper can identify the causal impact of becoming eligible for margin lending around the policy threshold. It can also document that the market anticipated much of that effect in the broader sample.

But it cannot cleanly identify the entire role of margin debt in the 2015 crash. By then, shadow margin was important and there is no sharp quasi-experimental variation through the aggregate peak and collapse.

Interpretation. The boom-bust calculations are informative and disciplined, but they should be read as suggestive rather than as the paper's cleanest causal result.

4.11 Relation to the literature

The paper sits at the intersection of three literatures:

- the literature on credit expansions and housing booms,
- the literature on margin regulation, which often studies volatility rather than price levels,
- and the literature on arbitrage, anticipation, and front-running in asset markets.

Its distinctive contribution is to show that a large stock-market credit expansion can causally raise prices, and that unconstrained investors need not offset the effect — they may amplify it by buying in advance.

5 Tables and figures

5.1 Table 1, Figure 1, and Figure 2: the scale and mechanics of the deregulation

Table 1 Number of marginable stocks by vintage				
Vintage	Announcement date	# newly marginable		% of floating cap
		Shanghai	Shenzhen	
Pilot A	February 13, 2010	50	40	53.05
Pilot B	November 25, 2011	131	60	65.86
1	January 25, 2013	163	113	75.94
2	September 6, 2013	104	102	78.22
3	September 12, 2014	104	114	79.30

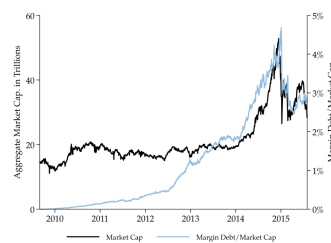


Figure 1
Aggregate market cap and margin debt/market cap over time
This plot shows daily aggregate floating market capitalization (in black) and the ratio of aggregate margin debt to aggregate floating market capitalization (in blue) for all stocks in our sample, including those that are never marginable. Floating market capitalization is measured in trillion of yuan.

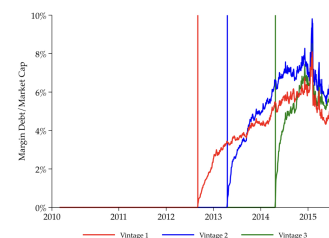


Figure 2
Margin debt/market cap by vintage
This plot shows the daily ratio of aggregate margin debt to aggregate floating market capitalization for the three vintages we study. Vertical lines denote starting dates of each vintage.

Read:

- Table 1 shows a clear stepwise expansion from pilot coverage of 53.05% of floating market cap to 79.30% by Vintage 3.
- Figure 1 shows aggregate formal margin debt rising from essentially zero to roughly 4.5% of floating market capitalization, alongside a large rise in total market capitalization.

- Figure 2 shows that within each vintage, margin debt reaches about 3% to 5% of floating market capitalization within a few months and peaks at roughly 8% to 10%.

Economic message. This is not a small policy tweak. The paper studies a genuinely large shock to leverage in the stock market.

5.2 Table 2: event-study evidence on anticipation

Read:

- Post-marginability abnormal returns are basically zero: 0.3% over 1 month, -0.3% over 3 months, and 1.7% over 12 months.
- Pre-marginability abnormal returns are large and increasing as the event approaches: 2.1% in the prior month, 7.3% in the prior 3 months, and 26.9% in the prior 12 months.
- The treated-minus-control difference from 12 months before to 12 months after is still about 25.3% .
- Placebo vintages generate no comparable pattern.

Economic message. The main treatment effect is already entering prices before the official event. That is the central empirical pattern the paper must explain.

Table 2
Event study of marginability

A. Cumulative returns before and after the introduction of margin lending

	Following marginability			Preceding marginability			Before vs. After
	0 to 1	0 to 3	0 to 12	-1 to 0	-3 to 0	-12 to 0	-12 to 12
Marginable	0.003	-0.003	0.017	0.021***	0.073***	0.269***	0.253***
	(0.004)	(0.008)	(0.016)	(0.005)	(0.009)	(0.015)	(0.023)
N	3543	3483	3146	3562	3562	3562	3146

B. Placebo estimates

	Following marginability			Preceding marginability			Before vs. After
	0 to 1	0 to 3	0 to 12	-1 to 0	-3 to 0	-12 to 0	-12 to 12
Marginable	-0.001	-0.002	-0.011	0.002	0.014	0.049	0.038
	[-0.013, 0.012]	[-0.023, 0.022]	[-0.053, 0.024]	[-0.008, 0.015]	[-0.034, 0.042]	[-0.024, 0.12]	[-0.058, 0.129]

In panel A, the first three columns show results from stock level regressions of cumulative DGTW adjusted returns from the month of marginability to 1 month, 3 months, or 12 months after on an indicator equal to one for newly marginable stocks. Columns 4-6 consider from 1, 3, and 12 months preceding the introduction to the month of the introduction itself. Column 7 considers the period from 12 months before to 12 months after the introduction. For each of the three vintages we consider a cross-section that includes the newly marginable stocks in that vintage as well as the set of never marginable stocks. Standard errors, clustered at the stock level, are included in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$. Panel B provides means from placebo replications of the analysis in panel A. For each of 1,000 placebo draws, we mimic the implementation of the reform. We first draw three random event months in the window between January 1993 and December 2008. At the first event month, we replicate the pilot vintages by selecting the top 150 stocks in each exchange. We then sequentially move through each of the three event dates, assigning the top 100 not yet-selected stocks in each exchange to vintages 1, 2, and 3. With these vintages defined, we conduct event studies at various horizons that compare the treated stocks to the set of stocks that are never treated. All draws are with replacement. Brackets contain empirical 95% confidence intervals, displaying the 2.5th and 97.5th percentiles of the distribution of placebo estimates. In both panels, all specifications include dummy variables for each vintage as controls.

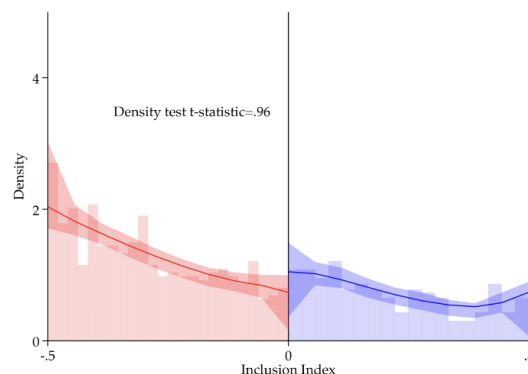


Figure 3
No evidence of bunching at threshold
This plot shows the results of the manipulation tests outlined in Cattaneo, Jansson, and Ma (2018). Red and blue bars show a histogram of the underlying data, while red and blue lines show local polynomial estimates of the density. We report the robust, bias-corrected t -statistic.

5.3 Figure 3: no bunching at the threshold

Read:

- The density of the running variable is smooth on both sides of the marginability threshold.
- The reported robust bias-corrected manipulation statistic is close to zero in economic terms and below conventional significance cutoffs.

Economic message. The threshold appears usable for causal inference. Investors and firms do not seem able to sort precisely around it.

5.4 Figure 4 and Table 3: the local causal effect of margin debt

Read:

- Panel A of Figure 4 and Panel A of Table 3 show an almost discontinuous jump in treatment at the threshold: marginability rises by about 0.959, stock-level margin debt rises by about 0.359 billion yuan, and margin debt / market cap rises by about 0.049.
- Panel B of Table 3 shows reduced-form cumulative returns of about 4.6% after 5 trading days, 8.3% after 20 trading days, and 24.6% after 60 trading days.
- Panel C of Table 3, the fuzzy RD, delivers almost identical causal estimates: 4.7%, 8.5%, and 26.0%.

Economic message. Margin eligibility causes a large positive stock-price effect. The event-study pre-trends are not spurious noise; they are anticipation of a real causal shock.

5.5 Figure 5 and Table 4: institutions buy before and sell after

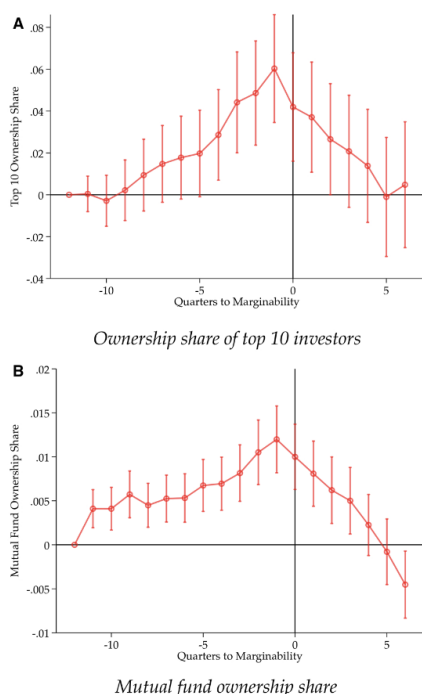


Figure 5

Institutional owners bought before the introduction of margin debt and sold afterwards

Coefficients from event-study regressions of ownership by institutions in the quarters surrounding the introduction of margin debt. For each vintage k , we construct a panel of all stocks that become marginable in k (treated stocks) and all never-marginable stocks from 12 quarters before the introduction to 6 quarters after. We stack these panels, and plot the coefficients from the following regression for stock j in vintage k and quarter t :

$$\text{Ownership Share}_{jkt} = \sum_{\ell} \beta_{\ell} \times \mathbb{1}\{t - \tau(k) = \ell\} \times \text{Treated}_{jk} + \gamma_{jk} + \lambda_{tk} + e_{jkt}$$

ℓ runs from -11 to 6 , with 12 quarters prior to the introduction acting as the omitted group. $\mathbb{1}\{t - \tau(k) = \ell\}$ is an indicator equal to one if the difference between quarter t and the implementation date $\tau(k)$ of vintage k is ℓ . The dependent variables are the stock-level share of ownership by the top 10 investors (panel A) or mutual funds (panel B). Bars represent 95% confidence intervals based on standard errors clustered at the stock level.

Table 4

Mutual fund portfolio weights: anticipatory buying and post-marginability selling

	(1)	(2)	(3)	(4)
Top Quintile $_{i,t} \times \text{Months Since Last Vintage}_t$	0.0034*** (0.0011)			
Top 150 $_{i,t} \times \text{Months Since Last Vintage}_t$		0.0027** (0.0011)		
$\log(\text{Index}_{i,t}) \times \text{Months Since Last Vintage}_t$			0.0016** (0.0007)	
Marginable $_{i,t} \times \text{Months Since Marginable}_{i,t}$				-0.0075*** (0.0027)
Controls	Yes	Yes	Yes	Yes
Stock Fixed Effect	Yes	Yes	Yes	Yes
Fund Fixed Effect	Yes	Yes	Yes	Yes
Time Fixed Effect	Yes	Yes	Yes	Yes
N	324,860	324,860	324,860	455,846
Adj. R^2	0.446	0.446	0.446	0.442

The first three columns of this table report the results of regressions following Equation 5. Top Quintile equals one if the stock is ranked in the top 20% among all non-marginable stocks based on the inclusion index, and zero otherwise. Top 150 equals one if the stock is ranked within the top 150 among all non-marginable stocks based on the inclusion index, and zero otherwise. $\log(\text{Index})$ equals the log of the inclusion index for non-marginable stocks and zero for marginable stocks. Months Since Last Vintage equals the number of months between fund reporting and the previous vintage. The fourth column shows results of regressions following Equation 6. Marginable equals one if stock i is marginable in month t , and zero otherwise. Months Since Marginable equals the number of months since stock i became marginable, as of month t . Controls refer to the percentile ranking of stock characteristics, including floating market capitalization, book-to-market, turnover rate, past month return, and past year return, and the interactions between Months Since Last Vintage and the percentile ranking of each stock characteristic. Fund, stock, and time fixed effects are included in all specifications. The sample period is from December 2011 to June 2014 in columns 1–3 and from December 2011 to June 2015 in column 4. Standard errors are clustered by fund and reported in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

Read:

- Figure 5 shows that mutual-fund ownership and top-10 investor ownership rise before margin-

ability, peak around the quarter just before the rollout, and then decline afterward.

- In Table 4, funds overweight likely-to-be-marginable stocks more as the next vintage approaches: the interaction coefficients are 0.0034 for the top-quintile definition, 0.0027 for the top-150 definition, and 0.0016 when using the log inclusion index.
- After a stock becomes marginable, the interaction coefficient is -0.0075 , indicating that funds unwind these positions over time.

Economic message. This is exactly the front-running pattern the paper’s story needs: institutions build positions ahead of the retail-credit shock, then sell out as margin buyers arrive.

5.6 Table 5: mutual funds earn alpha from the anticipatory strategy

Read:

- Mutual funds allocate about 12.98% of AUM to top-quintile TBM stocks and 12.41% to top-150 TBM stocks.
- TBM portfolios earn 1.646% raw return per month versus 1.089% for NTBM holdings; the long-TBM/short-NTBM strategy earns 0.556% per month.
- The abnormal performance survives market adjustment, CAPM alpha, and DGTW adjustment. Under the top-quintile definition, TBM market-adjusted return is 0.830%, CAPM alpha is 0.844%, and DGTW return is 0.546% per month.

Economic message. Institutions not only anticipated the reform — they profited from doing so.

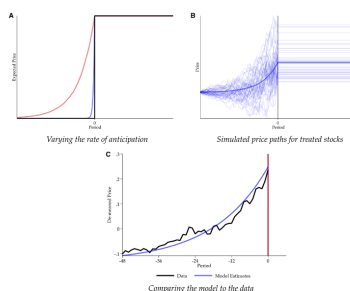


Figure 6
Anticipating an increase in credit supply

In all figures the y -axis displays price, the x -axis displays time, and the vertical line indicates the event date of the credit supply shock. Panel A shows the expected price path for stocks receiving a credit supply shock at time t (the vertical line). The black line shows the expected price path from a version of our model with no anticipation. The blue line shows the expected price path from a version of our model with a large value of θ . The red line shows the expected price path from a low value of θ . Panel B shows price realizations for treated stocks from simulations based on our model. Each lighter blue line represents the price path for an individual stock. The thick blue line represents the average price for all treated stocks in each period. Panel C plots, in black, average prices for the stocks in vignettes 1–3, after normalizing relative to the first month in our sample (March 2009) and realizing with respect to stock and month fixed effects (using the set of control stocks as a control group). In blue, the plot shows the (de-meaned) model implied price for the anticipated price increase.

[illegible]

A. Summary statistics		Crash return	Peak total
<i>All stocks</i>			
	Mean	-47.35%	6.69%
	Median	-46.58%	6.66%
	SD	26.49%	16.46%
<i>Sort on Peak Total Margin</i>			
Top Quintile	Mean	-41.54%	16.50%
	Median	-50.08%	13.35%
	SD	21.24%	12.52%
<i>Others</i>			
	Mean	-40.48%	3.71%
	Median	-44.82%	3.14%
	SD	24.17%	2.70%
<i>B. Regressions of crash returns on margin debt at the peak</i>			
		(1)	(2)
Peak Total Margin		-0.260** (0.116)	
Top Quintile/Peak Total Margin			-0.059** (0.014)
Controls	Yes		Yes
	1,645		1,645

Crash Return refers to the cumulative log return between the peak of margin debt on June 12 and November 27, 2015, a 24-week window capturing the crash period. *Peak Total Margin* refers to the level of formal and shadow margin debt relative to floating market capitalization on June 12, 2015. Panel A shows summary statistics for the best period. Panel B reports regression results of *Crash Return* on *Peak Total Margin* in column (1), and on an indicator equal to one if a stock is in the top quintile of *Peak Total Margin* in column (2). We control for stock characteristics including size, book-to-market, the past month return, the past year return, and turnover. Robust standard errors are included in parentheses. $^*p < .10$. $^{**}p < .05$. $^{***}p < .01$.

5.7 Figure 6 and Table 6: the dynamic model of anticipation fits the data

Read:

- Figure 6A shows that the model generates different pre-trend shapes depending on θ : small differences in θ change how early prices start rising.

- Figure 6C shows that the exponential path implied by the model tracks the average anticipatory price path in the data closely.
- Table 6 estimates θ between about 1.05 and 1.10 in the nonlinear models, and analogous values in the linear dynamic panel estimates.
- The implied direct effect ranges from roughly 20% to 39% depending on specification and sample. The preferred 2SLS base estimate implies a direct effect of about 39.3%, while the full-sample 2SLS estimate is about 20.0%.

Economic message. A simple information-revelation model goes a long way. It summarizes both the *size* of the credit shock and the *speed* with which markets learned about it.

5.8 Table 7: margin debt and the bust

Read:

- The average crash return from June 12 to November 27, 2015 is -42.35% .
- Mean peak total margin is 6.69% of floating market capitalization, with a top-quintile average of 16.55% .
- Stocks in the top quintile of peak total margin have an average crash return of -48.54% , versus -40.48% for the rest.
- In the cross-section, the coefficient on peak total margin is -0.266 , and the top-quintile indicator carries a -0.059 coefficient.

Economic message. The paper stops short of a strong causal claim here, but the association is consistent with the popular narrative that leverage intensified the crash.

5.9 Back-of-the-envelope boom decomposition

Read:

- Using the RD price effect of about 26% and the 5% increase in margin debt relative to market capitalization, the paper backs out an elasticity of demand of about -0.19 .
- Applying this to stock-level margin growth from June 2014 to June 2015, the mean implied boom return is 23.4% , versus an actual mean realized return of 112.7% .
- A cross-sectional regression of realized boom returns on implied boom returns yields a coefficient of 0.141 and an R^2 of 18.2% .

Economic message. Margin debt does not explain the whole boom, but it explains a meaningful piece of it.

6 Short summary

Hansman, Hong, Jiang, Liu, and Meng study the Chinese margin-lending deregulation of the early 2010s to ask whether a large credit expansion raises stock prices. Their reduced-form evidence

shows that treated stocks do not jump much on average after becoming marginable, but they do rise strongly beforehand. A regression discontinuity around the eligibility threshold reveals that this is not because credit has no effect; rather, the causal price effect is large — about 25% over 60 trading days — and much of it is anticipated. Mutual funds and other large investors buy likely-to-be-marginable stocks in advance, earn abnormal returns, and then sell after margin buyers arrive. A dynamic model with exponential information revelation rationalizes the pre-run-up and implies that most of the effect is priced in months before the policy date. The paper’s broader message is that large stock-market credit expansions can meaningfully raise prices and help fuel booms, even in the presence of sophisticated arbitrageurs.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes two connected facts. First, major increases in credit for equity purchases can cause stock prices to rise by a lot. Second, in liquid stock markets that effect may be hard to see in a conventional event window because sophisticated investors anticipate the shock and buy ahead of it. In the Chinese margin-lending episode, both statements are true: the causal effect is large, and the market prices much of it in advance.

7.2 Economic intuition

Credit expands the demand curve of constrained retail investors. That pushes prices up unless unconstrained investors are willing to absorb the extra demand at nearly unchanged prices. In China, the opposite happened. Unconstrained investors understood that new leverage would create future buying pressure, so they bought early rather than leaning against the shock. The resulting path of prices is therefore not a discrete jump on the event date, but a gradual anticipatory run-up followed by modest additional drift as the new credit is actually deployed.

7.3 Takeaway

This paper is a clean illustration of why “no post-event jump” does not mean “no treatment effect” in asset markets. If the policy is foreseeable, the right empirical design must separate anticipation from causality. Hansman et al. do that by combining broad event studies, a threshold-based RD, ownership data, and a parsimonious dynamic model. The main lesson is important well beyond China: shocks to leverage can move stock prices, and the arbitrageurs who understand that may amplify the boom rather than neutralize it.